

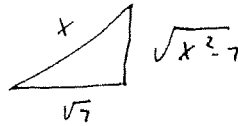
Worksheet

Evaluate the following integrals.

$$1. \int \frac{1}{\sqrt{x^2-7}} dx = \int \frac{1}{\sqrt{7} \tan \theta} \sqrt{7} \sec \theta \tan \theta d\theta = \int \sec \theta d\theta$$

$$x = \sqrt{7} \sec \theta$$

$$dx = \sqrt{7} \sec \theta \tan \theta d\theta$$



$$\ln |\sec \theta + \tan \theta| + C$$

$$\ln \left| \frac{x}{\sqrt{7}} + \frac{\sqrt{x^2-7}}{\sqrt{7}} \right| + C$$

$$2. \int \frac{1}{\sqrt{x^2+6x+34}} dx$$

Complete the square: $\int \frac{1}{\sqrt{(x+3)^2+25}} dx$

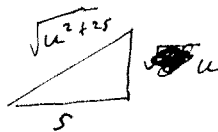
Let $u = x+3$
 $du = dx$

$$\int \frac{du}{\sqrt{u^2+25}} = \int \frac{1}{5 \sec \theta} 5 \sec^2 \theta d\theta = \int \sec \theta d\theta$$

$$\ln |\sec \theta + \tan \theta| + C$$

Let $u = 5 \tan \theta$

$$du = 5 \sec^2 \theta d\theta$$



$$\ln \left| \frac{\sqrt{u^2+25}}{5} + \frac{u}{5} \right| + C$$

$$3. \int \frac{3x+2}{(x-1)(x+4)} dx =$$

$$\ln \left| \frac{\sqrt{(x+3)^2+25}}{5} + \frac{x+3}{5} \right| + C$$

$$\int \frac{A}{x-1} + \frac{B}{x+4} dx = \int \frac{1}{x-1} + \frac{2}{x+4} dx = \ln |x-1| + 2 \ln |x+4| + C$$

$$A(x+4) + B(x-1) = 3x+2$$

Method I

$$(A+B)x + 4A - B = 3x + 2$$

$$\begin{aligned} A+B &= 3 \\ 4A-B &= 2 \end{aligned} \Rightarrow \begin{aligned} 5A &= 5 \Rightarrow A=1 \\ B &= 2 \end{aligned}$$

Method II

Let $x=1$ $5A=5 \Rightarrow A=1$

Let $x=-4$ $-5B=-10 \Rightarrow B=2$

$$4. \int \frac{x-1}{(x+1)(x^2+3x+4)} dx =$$

$$\int \frac{A}{x+1} + \frac{Bx+C}{x^2+3x+4} dx$$

$$A(x^2+3x+4) + (Bx+C)(x+1) = x-1$$

$$A(x^2+3x+4) + Bx(x+1) + C(x+1) = x-1$$

$$\underline{x=-1} \quad 2A = -2 \quad \Rightarrow A = -1$$

$$\underline{x=0} \quad 4A + C = -1 \quad \Rightarrow C = 3$$

$$\underline{x=1} \quad B = 1$$

$$\int \frac{-1}{x+1} + \frac{x+3}{x^2+3x+4} dx$$

$$- \ln|x+1| + \int \frac{x+3}{(x+\frac{3}{2})^2 + \frac{7}{4}} dx$$

$$\text{Let } u = x + \frac{3}{2} \\ du = dx$$

$$\int \frac{(u-\frac{3}{2})+3}{u^2 + \frac{7}{4}} du$$

$$\int \frac{u}{u^2 + \frac{7}{4}} du + \int \frac{\frac{3}{2}}{u^2 + \frac{7}{4}} du$$

$$\frac{1}{2} \ln|u^2 + \frac{7}{4}| + \frac{3}{2} \frac{1}{\sqrt{\frac{7}{4}}} \tan^{-1}\left(\frac{u}{\sqrt{\frac{7}{4}}}\right)$$

$$- \ln|x+1| + \frac{1}{2} \ln|(x-\frac{3}{2})^2 + \frac{7}{4}| + \frac{3}{2} \frac{1}{\sqrt{\frac{7}{4}}} \tan^{-1}\left(\frac{x-\frac{3}{2}}{\sqrt{\frac{7}{4}}}\right) + C$$

$$5. \int \frac{x^2+x-5}{x+3} dx =$$

$$\begin{array}{r} x-2+\frac{1}{x+3} \\ x+3 \overline{) x^2+x-5} \\ \underline{x^2+3x} \\ -2x-5 \\ \underline{-2x-6} \\ 1 \end{array}$$

$$\int x-2 + \frac{1}{x+3} dx$$

$$\frac{1}{2} x^2 - 2x + \ln|x+3| + C$$